

CE 310 — Week 6 Weekly Quiz

Week: 6 | **Topic:** Empirical Percentiles Across CE Practice

Due: Closes 8:00 AM Tuesday, Oct 6 — the day before Exercise 6 and Homework 6 are due. Taken on D2L

Points: 15 questions × 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — LARGE/SMALL/PERCENTILE, exceedance probability, duration curves, and how ACI 318, the 85th-percentile speed rule, and AACE contingency practice apply these tools. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. [Matching] Match each Excel formula to what it returns for a column of 500 cost records.

Choices:

- (A) =LARGE(range, 50)
- (B) =SMALL(range, 50)
- (C) =PERCENTILE(range, 0.10)
- (D) =COUNTIF(range, ">250")/500*100
- (E) =SUMPRODUCT((range-250)*(range>250))

Matches:

1. The percentage of records exceeding \$250
 2. The 50th-smallest record actually observed
 3. The total dollar amount by which above-threshold records exceed \$250
 4. The 10th percentile, interpolated between two adjacent sorted values
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Q2. The **exceedance probability** $P(X > x)$ is defined as:

- A) The probability that X equals exactly x
 - B) The fraction of records less than x
 - C) The fraction of records that exceed x
 - D) The number of standard deviations between x and the mean
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Q3. On a duration curve built by ranking a variable from highest to lowest, the x-axis represents:

- A) Time, in chronological order
 - B) Rank — 1 for the highest value, n for the lowest
 - C) The variable's own value at the same point in the record
 - D) Cumulative probability from 0 to 1
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Q4. ACI 318's concrete compliance rule (no more than 1-in-10 tests below spec) is fundamentally:

- A) A Normal-distribution assumption about concrete strength
 - B) A 10th-percentile compliance test, applied empirically to real test data
 - C) A requirement that the mean strength exceed the spec by 10%
 - D) Only valid when the sample size exceeds 1,000 observations in the full record
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Q5. Site noise readings are ranked to report standard acoustic descriptors. **L10** is defined as the level exceeded 10% of the time. Which percentile of the readings is it, and which function returns it?

- A) The 10th percentile — use SMALL, because the name says 10
 - B) The 10th-largest reading — use LARGE(range, 10), regardless of how many readings there are
 - C) The median — use MEDIAN, since L10 and L90 straddle it symmetrically
 - D) The 90th percentile — use LARGE, because a level exceeded only 10% of the time is high, not low
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Q6. [Ordering] Put the steps of building a duration curve in the order you would carry them out.

- (A) Sort the values from highest to lowest
 - (B) Assign each value its rank — 1 for the highest, n for the lowest
 - (C) Convert each rank into an exceedance probability, rank / n
 - (D) Plot value against exceedance probability, and read the design percentile off the curve
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Q7. [Select all that apply] Which of the following are true of the empirical approach — counting directly from the data — compared with fitting a parametric distribution?

- (A) It makes no assumption about the shape of the distribution
 - (B) It can report a value more extreme than anything in the dataset
 - (C) It is limited by sample size: 500 records cannot resolve a 1-in-1,000 event
 - (D) It requires the data to be approximately Normal
 - (E) It is the basis for ACI 318's 1-in-10 compliance rule
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Q8. Three standards in this week's material each set a threshold from a percentile: **ACI 318** uses the 10th percentile of concrete strength, the **85th-percentile speed rule** uses the 85th of observed speeds, and **AACE** contingency uses the 90th percentile of project cost. What decides whether a standard reaches for a low percentile or a high one?

- A) The sample size available — small samples force a percentile nearer the middle
 - B) Whether the underlying distribution is skewed to the left or to the right
 - C) Professional convention within each field; the choice carries no engineering meaning
 - D) The direction in which the variable causes harm — low where failure means too low
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Q9. Supplier Alpha has 243 test cylinders in the record. A colleague computes Alpha's 10th percentile of concrete strength using =SMALL(FILTER(D2:D501,B2:B501="Alpha"),24) and gets 4,181 psi, then computes it again using =PERCENTILE(FILTER(D2:D501,B2:B501="Alpha"),0.10) and gets 4,186 psi. Why do the two answers differ slightly?

- A) SMALL returns an observed value; PERCENTILE interpolates between two of them
 - B) One of the two formulas must contain an error
 - C) PERCENTILE only works on the full dataset, not a filtered subset
 - D) The two functions require different units
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Q10. [True/False] When an empirical percentile obtained by counting disagrees with one predicted by a fitted distribution, the empirical value is the more defensible basis for a compliance decision, because it assumes nothing about the shape of the distribution.

Q11. Percentiles can be computed two ways — LARGE/SMALL for an exact rank, and PERCENTILE for an interpolated value. When should you use the exact-rank method?

- A) Whenever the dataset holds fewer than 1,000 rows, since interpolation is unstable there
 - B) For code compliance — ACI 318 defines its rule as a fixed fraction of tests
 - C) Only when the target percentile lands exactly on a whole-numbered rank
 - D) Never; PERCENTILE interpolates and is therefore always more accurate
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Q12. For 500 sorted readings, which pair of formulas returns the same value?

- A) =MEDIAN(range) and =PERCENTILE(range, 0.50)
 - B) =MEDIAN(range) and =LARGE(range, 250)
 - C) =MEDIAN(range) and =AVERAGE(range)
 - D) =PERCENTILE(range, 50) and =SMALL(range, 250)
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Q13. You need the number of Roadside readings that are also above 65 dB, with Location in column B and level in column C. Which formula returns it?

- A) =COUNTIFS(B2:B501, "Roadside", C2:C501, ">65")
 - B) =COUNTIF(C2:C501, ">65")
 - C) =COUNTIF(B2:B501, "Roadside") + COUNTIF(C2:C501, ">65")
 - D) =COUNTIFS(C2:C501, ">65") divided by the number of Roadside rows
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Q14. You compare the empirical 10th-percentile strength against Week 5's fitted =NORM.INV(0.10, mean, sd) model, and the two disagree by about 90 psi. What does that tell you?

- A) The empirical value is wrong, because the fitted Normal model is the standard
 - B) The Normal model fits this tail less well than it fits the centre of the data
 - C) The dataset is too small for tail percentiles to carry any real meaning
 - D) NORM.INV should have been given 10 rather than 0.10 as its first argument
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Q15. FILTER is used for conditional percentiles — e.g. the 90th percentile of Roadside readings only. On an older Excel it returns #NAME?. What does that mean?

- A) The range contains text that must be cleaned out before FILTER will run
- B) The condition matched no rows, so there was nothing for FILTER to return
- C) FILTER does not exist at all in that version of Excel
- D) The workbook must be saved as .xlsx before FILTER becomes available